

Programmed death-1-induced interleukin-10 production by monocytes impairs CD4+ T cell activation during HIV infection

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Evidence abstract

Viral replication and microbial translocation from the gut to the blood during HIV infection lead to hyperimmune activation, which contributes to the decline in CD4+ T cell numbers during HIV infection.

Programmed death-1 (PD-1) and interleukin-10 (IL-10) are both upregulated during HIV infection.

Blocking interactions between PD-1 and programmed death ligand-1 (PD-L1) and between IL-10 and IL-10 receptor (IL-10R) results in viral clearance and improves T cell function in animal models of chronic viral infections.

Here we show that high amounts of microbial products and inflammatory cytokines in the plasma of HIV-infected subjects lead to upregulation of PD-1 expression on monocytes that correlates with high plasma concentrations of IL-10.

Triggering of PD-1 expressed on monocytes by PD-L1 expressed on various cell types induced IL-10 production and led to reversible CD4+ T cell dysfunction.

We describe a new function for PD-1 whereby microbial products inhibit T cell expansion and function by upregulating PD-1 levels and IL-10 production by monocytes after binding of PD-1 by PD-L1.

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